

EE 102 Week 6, Lecture 2 (Fall 2026)

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1 Introduction and Review

We start by reviewing concepts we have discussed in class before: (1) periodicity of signals, (2) properties of convolution, and (3) the eigenfunction property of complex exponentials for LTI systems.

1.1 Periodicity of signals

Recall the periodicity of signals by solving the popquiz below.

Pop Quiz 1.1: Check your understanding!

Prove that $x(t) = e^{j\omega t}$ is a periodic signal. Find its fundamental period.

Solution on page 7

As a follow-up, solve another popquiz! Once you're done, you can review the solutions at the end.

Pop Quiz 1.2: Check your understanding!

Prove that the following signals are period:

1. Scaled harmonic of the complex exponential: $x(t) = e^{jk\omega_0 t}$, where k is an integer.
2. Finite harmonic sum: $x(t) = \sum_{k=-K}^K a_k e^{jk\omega_0 t}$, where K is a positive integer.

Solution on page 7

1.2 Properties of convolution

We note two important properties of convolution that often come in handy: commutativity and associativity. For two signals $x(t)$ and $h(t)$, we have

$$x(t) * h(t) = h(t) * x(t),$$

How can we prove this? Hint: Use the change of variable (colloquially known as “u-substitution”).

To prove commutativity, we can start with the definition of convolution for the left-hand side:

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau.$$

Now, substitute $u = t - \tau$. Then, $du = -d\tau$ because t is a fixed value for the function inside the integral. The limits of integration become: $+\infty$ to $-\infty$ when τ goes from $-\infty$ to $+\infty$. Therefore, we have

$$x(t) * h(t) = \int_{+\infty}^{-\infty} x(t-u)h(u)(-du) = \int_{-\infty}^{+\infty} h(u)x(t-u)du = h(t) * x(t).$$

Associativity of convolution can be proved similarly. For three signals $x_1(t)$, $h(t)$, and $x_2(t)$, we have

$$x_1(t) * (h(t) * x_2(t)) = (x_1(t) * h(t)) * x_2(t).$$

To prove associativity, we can start with the left-hand side:

$$x_1(t) * (h(t) * x_2(t)) = \int_{-\infty}^{\infty} x_1(\tau) \left[\int_{-\infty}^{\infty} h(\lambda)x_2(t-\tau-\lambda) d\lambda \right] d\tau.$$

Now, we can change the order of integration:

$$x_1(t) * (h(t) * x_2(t)) = \int_{-\infty}^{\infty} h(\lambda) \left[\int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau-\lambda) d\tau \right] d\lambda.$$

The inner integral is the convolution of $x_1(t)$ and $x_2(t)$ evaluated at $t - \lambda$. Therefore, we have

$$x_1(t) * (h(t) * x_2(t)) = \int_{-\infty}^{\infty} h(\lambda)(x_1 * x_2)(t - \lambda) d\lambda = (x_1 * x_2) * h(t).$$

1.3 Eigenfunctions of LTI systems

At the end of the previous lecture, we established that $e^{j\omega t}$ is an eigenfunction of LTI systems. That is, for an input $x(t) = e^{j\omega t}$, the output $y(t)$ is also a complex exponential at the same frequency ω but scaled by a complex number $H(j\omega)$:

$$y(t) = H(j\omega)e^{j\omega t}.$$

where

$$H(j\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau.$$

As a result of this *very important* result, we proposed that if we are able to write any signal $x(t)$ as a linear combination of complex exponentials, then we can find the output $y(t)$ of an LTI system by simply scaling each complex exponential by $H(j\omega)$ and adding them up!

The previous line is a one-line summary of Fourier analysis and synthesis — something that we will be spending a lot of time on in the next few weeks.

We start this journey by posing the following problem: Given a T -periodic signal $x(t)$, can we write it as a linear combination of complex exponentials? If so, how? More concretely, can we write

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T},$$

the complex Fourier series coefficients are $\{a_k\}$. Note that $a_k \in \mathbb{C}$ are complex numbers, in general. The big question is; how do we find a_k ?

2 Goals

Represent a periodic signal $x(t)$ as a linear combination of harmonically related complex exponentials:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}. \quad (1)$$

Find the coefficients a_k .

3 Introduction to Fourier Series

Since the first part of the goal is “any periodic signal $x(t)$ ” and we are claiming that $x(t)$ can be written as a linear combination of complex exponentials. So, it must be true that (and we should make sure of it) that $e^{jk\omega_0 t}$ is periodic for every integer k .

Let’s start to answer the second part of the goal: how do we find a_k ? We will start with a simple example.

3.1 Example: A mix-sinusoidal audio signal

Use an idealized combination of three tones to see how Fourier coefficients encode frequency, amplitude, and phase. Real musical notes usually contain several harmonics; the example below is a simple sum of sinusoids.

Consider

$$x(t) = \sin(6t) + \cos(2t) + \sin(12t), \quad \omega_0 = 2.$$

Write $x(t)$ as a linear combination of $e^{jk\omega_0 t}$:

$$\begin{aligned}\sin(6t) &= \frac{1}{2j}(e^{j6t} - e^{-j6t}) = \frac{1}{2j}(e^{j(3)\omega_0 t} - e^{-j(3)\omega_0 t}), \\ \cos(2t) &= \frac{1}{2}(e^{j2t} + e^{-j2t}) = \frac{1}{2}(e^{j(1)\omega_0 t} + e^{-j(1)\omega_0 t}), \\ \sin(12t) &= \frac{1}{2j}(e^{j12t} - e^{-j12t}) = \frac{1}{2j}(e^{j(6)\omega_0 t} - e^{-j(6)\omega_0 t}).\end{aligned}$$

Hence

$$x(t) = \sum_{k=-6}^6 a_k e^{jk\omega_0 t}, \quad \omega_0 = 2,$$

with the nonzero Fourier series coefficients as

$$a_{\pm 1} = \frac{1}{2}, \quad a_{\pm 3} = \pm \frac{1}{2j}, \quad a_{\pm 6} = \pm \frac{1}{2j},$$

where the “ $\pm$ ” pairs obey $a_{-k} = a_k^*$ for this real $x(t)$.

The cosine contributes equal real coefficients at $k = \pm 1$. The sine terms contribute opposite, purely imaginary coefficients at $k = \pm 3, \pm 6$. All other a_k are zero. The fundamental period of this signal is $T = \pi$.

4 The Fourier analysis equation

To find a_k generally, we have the Fourier series analysis equation (the equation that gives us values of a_k):

$$a_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt \quad n \in \mathbb{Z},$$

if you replace n by k , you get the more familiar form:

$$a_k = \frac{1}{T} \int_0^T x(t) e^{-jk\omega_0 t} dt, \quad k \in \mathbb{Z},$$

which is the formula for the Fourier series coefficients. Note that for $k = 0$, we have

$$a_0 = \frac{1}{T} \int_0^T x(t) dt,$$

which is the average value (or DC component) of $x(t)$ over one period.

5 Fourier series properties

There are many helpful properties that you should know about Fourier series coefficients. Here are a few of them (you can find the full list in the textbook):

5.1 Linearity

For two signals represented using the same period T and harmonic frequencies, with Fourier series coefficients a_k and b_k , if we construct another signal by linear superposition, $z(t) = Ax(t) + By(t)$, then the Fourier Series coefficients for $z(t)$ satisfy

$$z(t) \iff \{Aa_k + Bb_k\}_{k \in \mathbb{Z}}.$$

We will talk about many more properties next week.

6 Practice Problems

1. Review continuous-time Fourier series analysis, synthesis, and properties in Chapter 3 of Oppenheim and Willsky, *Signals and Systems* (2nd edition), especially Sections 3.2–3.5 and Table 3.1.
2. Practice finding Fourier coefficients by integrating over one period of a rectangular waveform and a periodic ramp.
3. Use the LTI filtering rule to connect input and output Fourier coefficients, as required in Pset 6.

Reminder: Pset 6 is due October 14, 2026.

For an additional reference on harmonic analysis and LTI filtering, see [MIT 6.003 Lecture 15: Fourier Series](#).

Pop Quiz Solutions

Pop Quiz 1.1: Solution(s)

Periodicity requires us to prove that there exists a $T > 0$ such that $x(t + T) = x(t)$ for all t . We have

$$x(t + T) = e^{j\omega(t+T)} = e^{j\omega t} e^{j\omega T} = x(t) e^{j\omega T}.$$

Therefore, we need to find a $T > 0$ such that $e^{j\omega T} = 1$. This is true if and only if $\omega T = 2\pi k$ for some integer $k > 0$. Therefore, we can choose $T = 2\pi k/\omega$. The fundamental period is the smallest positive T , which occurs when $k = 1$, so the fundamental period is $T = 2\pi/\omega$.

Pop Quiz 1.2: Solution(s)

For $x(t) = e^{jk\omega_0 t}$, we have

$$x(t + T) = e^{jk\omega_0(t+T)} = e^{jk\omega_0 t} e^{jk\omega_0 T} = x(t) e^{jk\omega_0 T}.$$

Therefore, we need to find a $T > 0$ such that $e^{jk\omega_0 T} = 1$. This is true if and only if $k\omega_0 T = 2\pi m$ for some integer $m > 0$. Therefore, we can choose $T = 2\pi m/(k\omega_0)$. The fundamental period is the smallest positive T , which occurs when $m = 1$, so the fundamental period is $T = 2\pi/(k\omega_0)$. The second signal (summed harmonics) is also periodic with the same fundamental period $T = 2\pi/\omega_0$ since each term in the sum is periodic with that period, so the sum is also periodic with the same period.